

Markdowns under Factor-Augmenting Technology

Evidence from Indian Manufacturing, 2012–2022

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Technical summary — companion to the 40-page working paper

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1. The problem

Production with two productivity terms — Hicks-neutral Ω and **labour-augmenting** A :

$$Q_{ft} = G(A_{ft}L_{ft}, M_{ft}, K_{ft}; \beta) \cdot \Omega_{ft}$$

The literature’s two identities (De Loecker–Warzynski; De Loecker–Eeckhout–Unger), with $\psi^m = 1$ for competitive materials:

$$\mu_{ft}^p = \frac{\theta_{ft}^m}{\alpha_{ft}^m \psi_{ft}^m}, \quad \psi_{ft}^l = \frac{\theta_{ft}^l}{\theta_{ft}^m} \cdot \frac{\alpha_{ft}^m}{\alpha_{ft}^l} \cdot \psi_{ft}^m, \quad \mu_{ft}^w = \frac{\psi_{ft}^l - 1}{\psi_{ft}^l}$$

The second *looks* like a formula for ψ^l in observables. It is not. Once A_{ft} varies, θ^l and θ^m are *functions of* A_{ft} : a plant with more labour-augmenting productivity has more labour in effective units, hence a different marginal-product ratio. Unknowns sit on both sides.

	Assumption	Reads α^l/α^m variation as
Hicks-neutral (working paper 1)	$A_{ft} = A$ constant	the markdown ψ^l
Factor-biased (working paper 4)	$\psi^l = 1$	technology A_{ft}

Same moment, two labels. **Rubens, Wu and Xu (2025) prove neither is identified without an outside restriction** — so working papers 1 and 4 of this series cannot both be taken at face value. Their remedy: identify the inverse labour-supply elasticity from *labour supply*, outside the production function.

2. Method: three stages

Stage 1 — labour supply (breaks the tie)

Plain logit over state $\times$ year markets; outside option is not working in registered manufacturing.

$$\ln S_{ft} - \ln S_{0t} = \gamma W_{ft} + b \cdot \text{expshare}_{f,t-1} + \text{FE} + \xi_{ft}, \quad \boxed{\psi_{ft}^l = 1 + \frac{1}{\gamma W_{ft}(1 - S_{ft})}}$$

The nest was tested and *rejected at its boundary*: $\varsigma = -0.114$ (two-digit), -0.010 (five-digit), $+0.045$ (industry-only). Rubens–Wu–Xu find -0.019 , insignificant, and reach the same specification. *Instruments*: lagged plant export share $\times$ REER, and $\times$ industry tariff — shift-share, with the lag making exposure predetermined and the main effect controlled.

Stage 2 — input-ratio GMM

With $x = \ln(\alpha^l \psi^l / \alpha^m)$, $u = m - l$ and $v = u + x$ — note $v = w^l + \ln \psi^l - w^m$, the *markdown-adjusted relative input price*:

$$u_{ft} = c + \sigma v_{ft} + (1 - \sigma) a_{ft}$$

The AR(1) law of motion on $(1 - \sigma)a$ makes the innovation **bilinear** in the two parameters:

$$v_{ft}^a = u_t - \rho^a u_{t-1} - \sigma v_t + \sigma \rho^a v_{t-1}$$

so ρ^a concentrates out in closed form at each σ , and the profiled objective is *simultaneously* the estimator and the Stock–Wright weak-identification-robust statistic. Moments: $\mathbb{E}\left[v^a \mid w_{t-1}^l, w_{\ell t}^{\min}, w_{\ell, t-1}^{\min}\right] = 0$.

The normalisation — the paper’s key theoretical step

A is identified only relative to β^m . Writing $a = \tilde{a} + \kappa$ with $\kappa = -\sigma \ln \beta^m / (1 - \sigma)$ and $k = (\sigma - 1)/\sigma$, the labour term picks up $e^{k\kappa}$, and

$$k\kappa = \frac{\sigma - 1}{\sigma} \cdot \frac{-\sigma \ln \beta^m}{1 - \sigma} = \ln \beta^m \quad \textbf{exactly.}$$

So β^m multiplies *both* the labour and the materials term and factors out of the aggregator. Only $b = \beta^k / \beta^m$ survives:

$$\tilde{D}_{ft} = e^{k(\tilde{a}_{ft} + l_{ft})} + M_{ft}^k + b K_{ft}^k, \quad q_{ft} = \frac{\nu}{k} \ln \tilde{D}_{ft} + \text{const} + \omega_{ft}$$

Then, using $k\tilde{a} = -u/\sigma + v$ and $l - m = -u$,

$$\ln \frac{\theta_{ft}^l}{\theta_{ft}^m} = k(\tilde{a}_{ft} + l_{ft}) - k m_{ft} = v_{ft} - u_{ft} = x_{ft} \quad \textbf{identically, for every } \sigma.$$

Consequence. The wage markdown ψ^l , the markdown rate μ^w , and the entire comparison with working paper 1 are **invariant to the production parameters**. Only A_{ft} , the individual elasticities $\theta^l, \theta^m, \theta^k$, the returns to scale ν and hence the markup μ^p depend on σ . This is what makes the weak identification of σ (Section 6) an inconvenience rather than a threat.

Stage 3 — production GMM

The same bilinear form in $(\nu/k, \rho^\omega)$, on Akerberg–Caves–Frazer timing moments $\mathbb{E}[v^\omega \mid l_{t-1}, m_{t-1}, k_t, k_{t-1}] = 0$ (capital is chosen a period ahead, so k_t is predetermined). Elasticities follow directly:

$$\theta^l = \nu \frac{e^{k(\tilde{a}+l)}}{\tilde{D}}, \quad \theta^m = \nu \frac{M^k}{\tilde{D}}, \quad \theta^k = \nu \frac{b K^k}{\tilde{D}}, \quad \theta^l + \theta^m + \theta^k = \nu.$$

3. Data

423,127 plant-years, ASI 2012–2022, 36 states, 24 NIC-2 divisions, 377 state-year markets. Median plant: 89 persons, Rs 182,802 per worker. Screens: wage trim $0.5/99.5 \rightarrow 418,895$; market screen (manufacturing employment over working-age population ≤ 0.10 , removing the Daman & Diu and Dadra & Nagar Haveli migrant enclaves) $\rightarrow 413,438$.

Three inputs were constructed for this paper:

Input	Source	Result
Export share	raw ASI Block A/B, layout-driven parser	719,496 factory-years, 7.4% exporters; all eleven waves reconcile exactly
Minimum wage	Labour Bureau, <i>Report on the Working of the Minimum Wages Act, 1948</i> , Table 4	321 state-years, 2012–2020 ; 19.6% carried-forward and flagged; retains 30% of log variation after state and year effects, against 9.8% for the tariff
Working-age population	NCP <i>Population Projections 2011–2036</i>	no missing state-years

Two parser traps were found only by inspecting raw bytes against the official code lists: field **B03** is “**type of ownership**” only in **2012–14** (from 2015-16 the slot holds a masked 21-character CIN, returning 10²¹), and the **B02 code list breaks in 2015-16** — B02 is *legal form* while B03 is *ownership*, so a wholly government-owned firm registered as a public limited company sits in B02= 4.

Not available: no district field in any wave; the only sub-state identifier is the NSSO regional-office code. **State is the finest labour market obtainable**, against Rubens–Wu–Xu’s counties. This is the paper’s most important data limitation.

4. Estimates

Stage 1. $\hat{\gamma} = 0.4441$ (s.e. 0.1430), $t = 3.11$, $N = 290,834$, 65,001 plant clusters, $F = 13.0$, Hansen $J = 0.103$ (d.f. 1) $p = 0.749$.

Plant fixed effects are the binding choice. Without them γ changes sign across instrument sets (−1.20, +0.12, +0.13) and the Hansen test rejects ($p = 0.010, 0.033$) — a persistent plant amenity correlated with export status. With them $\gamma \approx 0.43$ everywhere and Hansen passes ($p = 0.75\text{--}0.94$). Instrument strength is comparable either way (F 5.9–20.3 without, 5.3–22.4 with), so there is **no strength-for-consistency trade-off**.

Anderson–Rubin 95% set: $\gamma \in [0.195, 1.030] \Rightarrow \text{median } \psi^l \in [1.48, 3.55] \Rightarrow \text{markdown } \mathbf{33\text{--}72\%}$

Stage 2. $\hat{\sigma} = 2.080$ (s.e. 0.173), $\rho^a = 0.525$, $J = 0.85$, $p = 0.356$. State effects are *forced by the test*: with year effects only the model is rejected, because minimum-wage **levels** are higher in richer states, which also differ permanently in labour-augmenting technology, so the quasi-difference leaves a state component the instrument is correlated with. Adding capital ($J = 29.1$) or second lags ($J = 124.5$) rejects overwhelmingly — the AR(1) is an approximation, so those moments are reported and **dropped**, not kept for the precision they would appear to buy.

Stage 3. Pooled: $\nu = 1.110$, $\rho^\omega = 0.886$, $b = 0.082$ — but $J = 88.8$, which **rejects**. Diagnosed rather than accepted: estimated separately by NIC-2 division (as working paper 1 estimates its Cobb-Douglas), the test rejects in only **3 of 23** divisions, median $p = 0.714$, ν median 1.085, ρ^ω median 0.845. Per-industry is therefore the preferred specification. Dropping each moment in turn moves ν only within $[0.905, 1.155]$.

5. Results

	p25	median	p75
θ^l CES (labour-augmenting)	0.130	0.243	0.408
θ^l Cobb-Douglas (working paper 1)	0.097	0.100	0.126
θ^m CES / Cobb-Douglas	0.607 / 0.876	0.792 / 0.902	0.916 / 0.933
Markup μ^p CES / Cobb-Douglas	0.888 / 1.058	1.055 / 1.209	1.178 / 1.476
ψ^l labour supply	1.722	2.120	2.791
ψ_{CD} Hicks-neutral (working paper 1)	0.376	0.825	1.814

(a) **The markdown is large, and it is not working paper 1’s.** Median 53% (weighted 55%). The Hicks-neutral estimate has a median $\psi_{CD} = 0.83$ — *below one*, implying the median plant pays *above* marginal revenue product; only **45.7%** of factories have $\psi_{CD} > 1$ at all. Plant-level correlation between the two series: **Pearson 0.0146, Spearman 0.0348**. They are not noisy measurements of one object; they are different objects.

(b) **The decomposition is exact.** Both estimators evaluate the same identity, so the entire difference is the elasticity ratio:

$$\ln \psi_{CD} - \ln \psi^l = \ln \frac{\theta_{CD}^l}{\theta_{CD}^m} - \ln \frac{\theta_{CES}^l}{\theta_{CES}^m}$$

The Cobb-Douglas ratio is constant within an industry by construction; the CES one moves with A . Standard deviation of θ^l : **0.234 (CES) against 0.041 (CD) — a factor of 5.7**. Regressing the gap on $(\tilde{a}, u)$ within NIC-2 returns $(-0.519231, +0.519231)$ against the theoretical $(-k, +k) = (-0.519231, +0.519231)$, a deviation of 2.1×10^{-15} . **The discrepancy is labour-augmenting productivity**, as an identity rather than an approximation.

(c) The bias reverses the Chinese result. Median log-gap -0.989 : the Hicks-neutral markdown is a factor **0.372** of the true one. Rubens–Wu–Xu find the opposite in Chinese nonferrous metals — 55% against a true 27%, roughly double, plus a manufactured 35%→73% trend where the true series is flat near 27%. **The sign of the bias is not a property of the estimator**: it is the covariance between labour-augmenting productivity and the labour cost share, which differs across settings. A Hicks-neutral markdown estimate is *not* conservative in either direction.

Aggregates. Employment-weighted markdown **0.461** — larger plants mark wages down *less* — against a plant-weighted 0.552. Plants with above-median markdowns account for **26.8%** of the wage bill.

Ownership (2017–2022, the clean code window): government $\psi^l = 1.598$ (37.4% markdown) < private corporate 1.827 < co-operative 1.879 < proprietary and partnership 2.618 (61.8%). Government plants also have the highest labour-augmenting productivity. The ordering is the same under both estimators, and agrees in direction with Rubens–Wu–Xu, who find markdowns 13% lower at Chinese state-owned enterprises. **Exporters** 1.960 against **non-exporters** 2.279.

6. Validation

Monte Carlo, reproducing the Rubens–Wu–Xu (IJIO) design: 50 markets $\times$ 5 plants $\times$ 10 years, Cobb-Douglas with $\beta^m = 0.30$, AR(1) productivity, logit labour supply, Nash–Bertrand wage competition. *Convention*: that design writes ψ for the inverse residual elasticity itself, with $\mu^w = \psi/(1 + \psi)$; the paper writes $\psi^l = 1 + \psi$. Same markdown either way.

	truth β^l	Hicks-neutral estimator	RWX joint estimator
DGP 1, Hicks-neutral	0.500	0.492	0.501
DGP 2, factor-biased	0.501 (sd 0.094)	0.085 , β^m 0.935, $\psi = -0.913$	0.501 (sd 0.094), β^m 0.300

Under factor bias the production-approach estimator collapses: it loads the difference onto materials and returns a *negative* inverse supply elasticity, an economically meaningless markdown. The joint estimator recovers the truth in both worlds *including the dispersion* of β^l — the object a Hicks-neutral model cannot represent at all.

Two identities asserted at run time (the pipeline stops if either fails): $\max |\ln \psi_{CES} - \ln \psi^l| = 4.8 \times 10^{-15}$ over 289,760 plant-years at four values of σ ; and the $(-k, +k)$ loading above.

Firm-cluster bootstrap — resamples plants and re-runs the *entire* chain $\gamma \rightarrow \psi^l \rightarrow (\sigma, \rho^a) \rightarrow (b, \nu, \rho^w) \rightarrow$ elasticities, markups, markdowns and the gap, so Stage-1 uncertainty propagates. 200 draws, 200 succeeded.

	95% percentile interval
γ	[0.224, 0.758] — inside the Anderson–Rubin set
median markdown	[0.396, 0.689] — against the AR-set range 33–72%
median log-gap	[−1.400, −0.733] — excludes zero; 100% of draws negative
σ	[0.05, 6.00] — the entire grid

7. What the paper refuses to claim

σ is only weakly identified. The Stock–Wright set is *unbounded* on $[0.05, 6]$: the profiled objective never exceeds 1.83 against a 5% critical value of 5.99, and the bootstrap independently spans the whole grid. The diagnosis is specific — after the state and year effects the overidentification test *forces*, $\ln w_t^{\min}$ and $\ln w_{t-1}^{\min}$ are correlated at **0.558**, so three moments span barely more than the two directions two parameters need; the first-stage F of the two minimum-wage instruments for v is 7.5. With 33 states and eight usable years this is a data limitation, not a fixable specification error. (Rubens–Wu–Xu’s own $\sigma = 0.397$ carries a standard error of 0.215.) **The invariance result of Section 2 is why this does not touch the headline:** the markdown is identical across $\sigma \in [0.4, 3.0]$ while every production-side object moves.

The cross-sectional and time patterns in ψ^l are largely mechanical. Under a single γ , $\psi^l = 1 + 1/(\gamma W(1 - S))$ is a decreasing function of the plant’s own wage, so any ranking of plants, states, industries or ownership types reproduces the inverse wage ranking — measured **rank correlation** -1.000 across states, industries *and* ownership. “Markdowns are highest in Tripura and Bihar” is the statement that wages are lowest there. The same mechanism drives the apparent decline over 2013–2022. Rubens–Wu–Xu allow a time-varying γ_t ; our instruments cannot support one. The log-gap does *not* inherit the mechanism (rank correlation with the wage 0.42 across states, 0.05 across industries) and is what the paper interprets.

Also stated: no plant-level output price, so the De Loecker–Goldberg–Khandelwal–Pavcnik critique applies to ω , ν and μ^p (though not to the markdown); minimum wages end in 2020, so Stage 2 runs 2013–2020; markets are states, not commuting zones.

8. Implication for the series

Working paper 1’s *conclusion* — that Indian manufacturing workers are paid materially below marginal revenue product — survives and strengthens: the markdown is larger under the labour-supply estimator, not smaller. Working paper 1’s *cross-section* does not survive: its plant-level ranking of who exercises labour market power is uncorrelated with the ranking obtained once labour-augmenting technology is admitted. **Working paper 2’s misallocation decomposition used those plant-level Hicks-neutral markdowns**, so that calculation does not survive unchanged. Redoing it follows directly from these estimates and is the natural next piece of work.

Reproducibility. Every figure in this brief traces to a saved result CSV or log. The method record is `Docs/MASTER.PLAN.md`; the result-to-code map is the `PAPER.PROVENANCE.GUIDE`; `Python/13.verify_paper.py` re-derives 92 quantitative claims of the full paper from the saved results and fails on any mismatch (current status: 92 checks, 0 mismatches). Plant-level ASI microdata are licensed by MOSPI and are not redistributable; all aggregates, code and the newly assembled minimum-wage series are shareable.